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Supplemental information

Unraveling the landscapes and regulation of scanning, leaky scanning, and 48S initiation complex conformations

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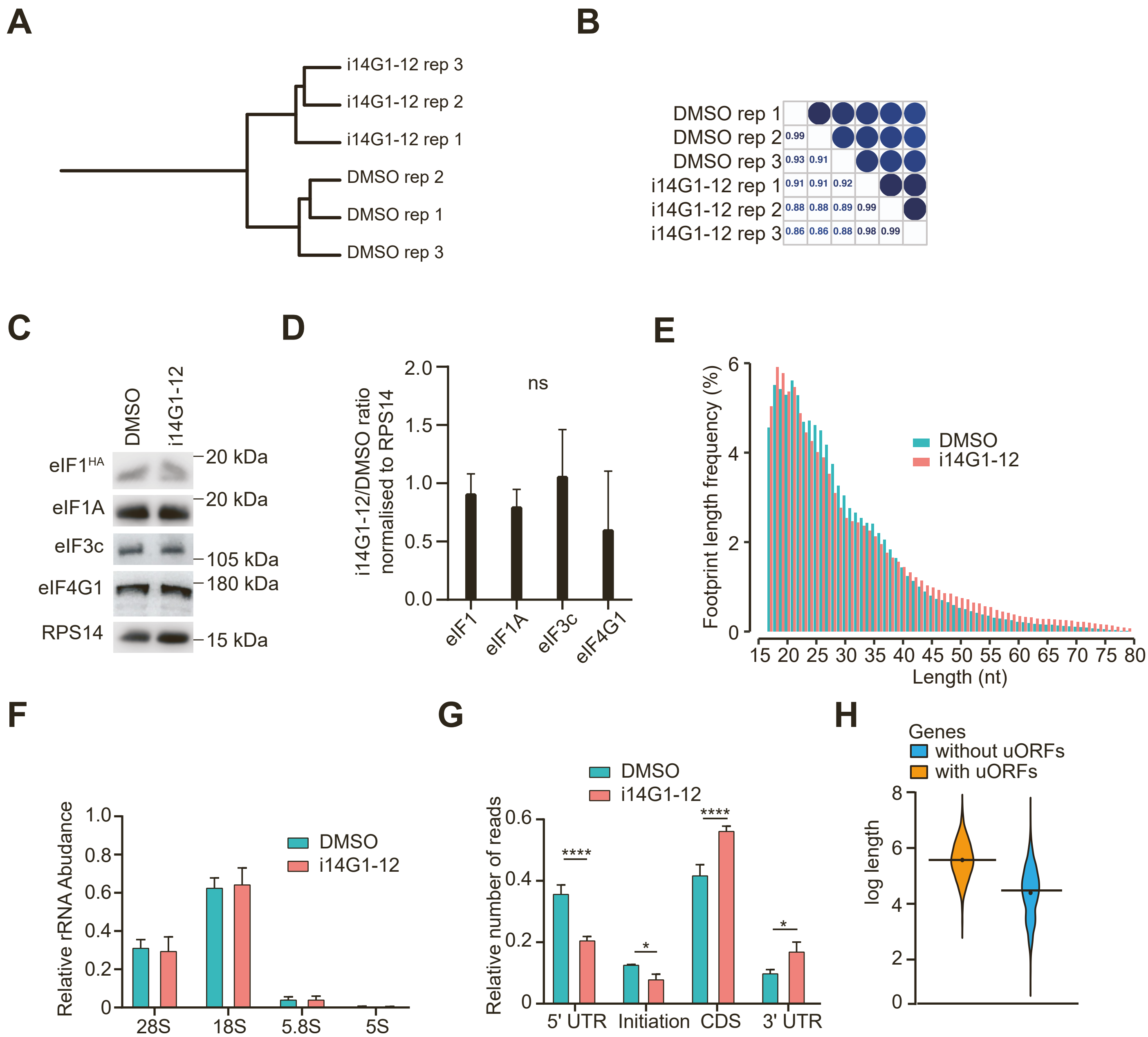

Figure S1: 40S TCP-seq quality control, Related to Figure 1

A-B. Phytogenic correlation tree **(A)** and correlation matrix **(B)** of the 40S TCP-seq samples.

C. Western blots of eIF1^{HA}, eIF1A, eIF3c, eIF4G1 or RPS14 from 40S fractions (Figure 1B). HEK293T treated with DMSO or i14G1-12 were processed through 40S TCP-seq. Fractions corresponding to the 40S were pulled and analysed by western blots.

D. Quantification of (C) from three biological replicates. The ratio of the protein abundance in i14G1-12 over DMSO was calculated and normalised to RPS19. The mean is reported ± SD (n = 3).

E. Histogram of the frequency of the footprints length in nucleotide from the 40S TCP-seq libraries. The mean is represented for each individual length.

F. Histogram of the relative rRNA abundance in DMSO or i14G1-12 treated cells. The mean is reported ± SEM.

G. Histogram of the relative number of reads mapping to several mRNA features (5' UTR, Initiation site, CDS or 3' UTR) from the 40S TCP-seq libraries. The mean is reported ± SEM (n = 3).

H. Violin plot of the log transformation of the 5' UTR length of genes containing or not upstream ORFs ; (•) mean, (—) median.

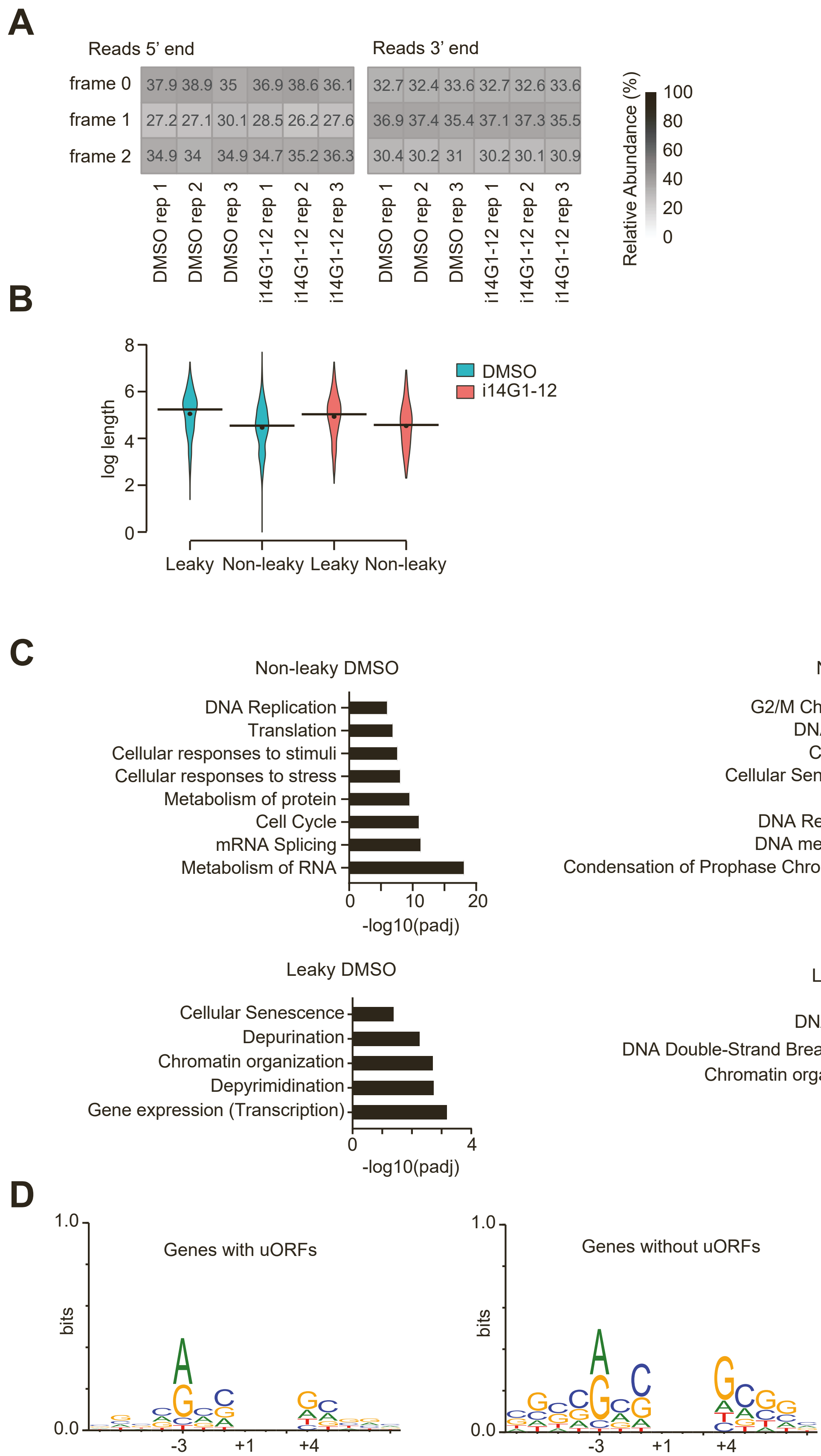

Figure S2: Leaky scanning analyses under i14G1-12, Related to Figure 2

A. Hetatmap of the frame usage of the reads ends mapping to the CDS.

B. Violin plot of the log transformation of the 5' UTR length of leaky or non leaky genes in DMSO or i14G1-12 ; (•) mean, (—) median.

C. Histograms of the most significantly enriched reactome terms of the significantly leaky (LS > 3) or non-leaky genes (LS < -3) in DMSO or in i14G1-12.

D. Weblogos of genes with (left) or without (right) experimentally confirmed upstream ORFs.

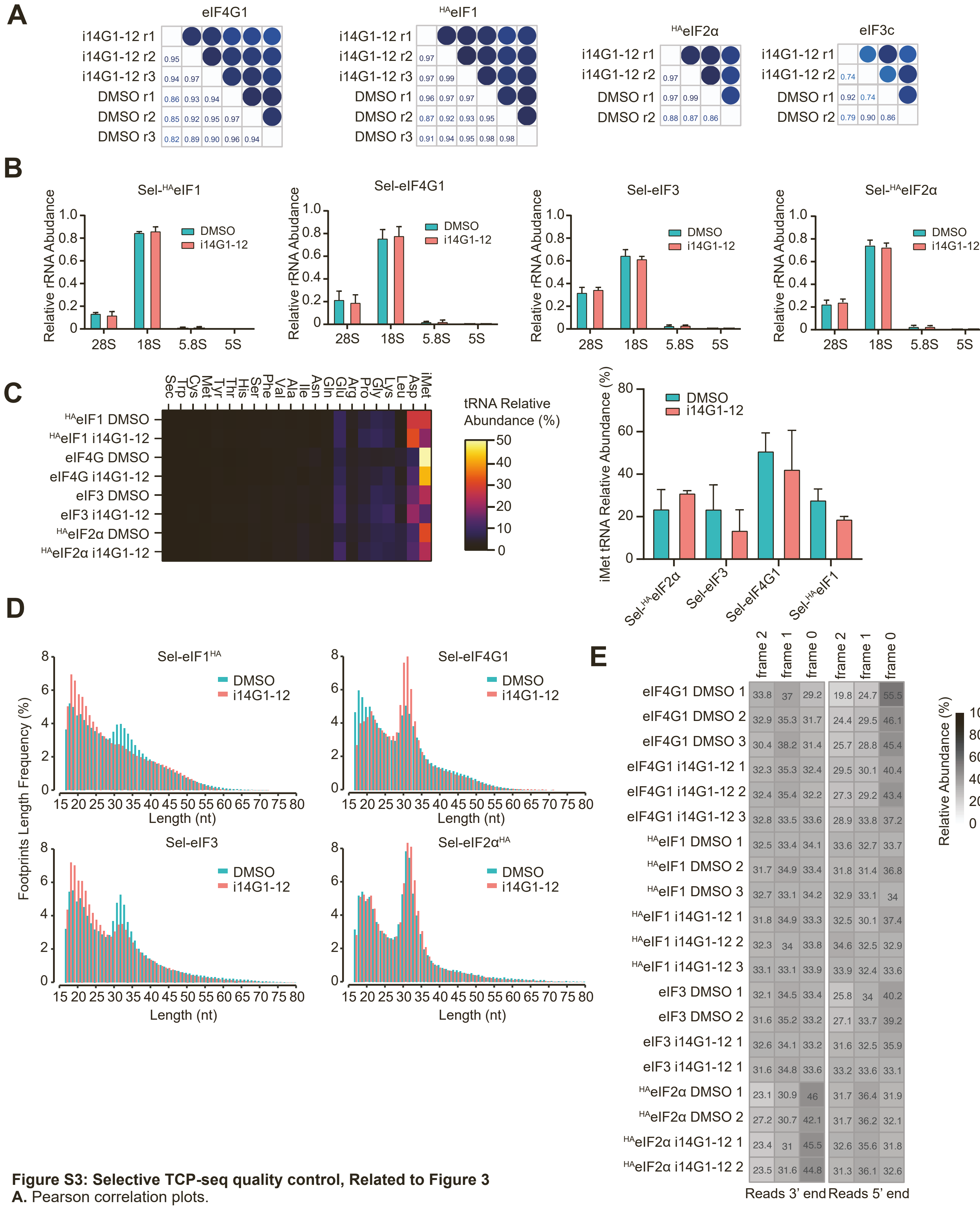

Figure S3: Selective TCP-seq quality control, Related to Figure 3

A. Pearson correlation plots.

B. Histograms of the relative rRNA abundance from the selective TCP-seq against eIF1^{HA}, eIF4G1, eIF3 or eIF2α^{HA}. The mean is reported ± SEM (n = 3 for eIF1 and eIF4G1; n = 2 for eIF3 and eIF2).

C. Heatmap of the mean relative tRNA abundance from the selective TCP-seq against eIF1^{HA}, eIF4G1, eIF3 or eIF2α^{HA} (left). Histogram of the abundance of the iMet tRNA (right). The mean is reported ± SEM (n = 3 for eIF1 and eIF4G1; n = 2 for eIF3 and eIF2).

D. Reads length distribution of the selective TCP-seq data libraries against eIF1^{HA}, eIF4G1, eIF3 or eIF2α^{HA}.

E. Heatmap of the frame usage of the reads ends mapping to the CDS.

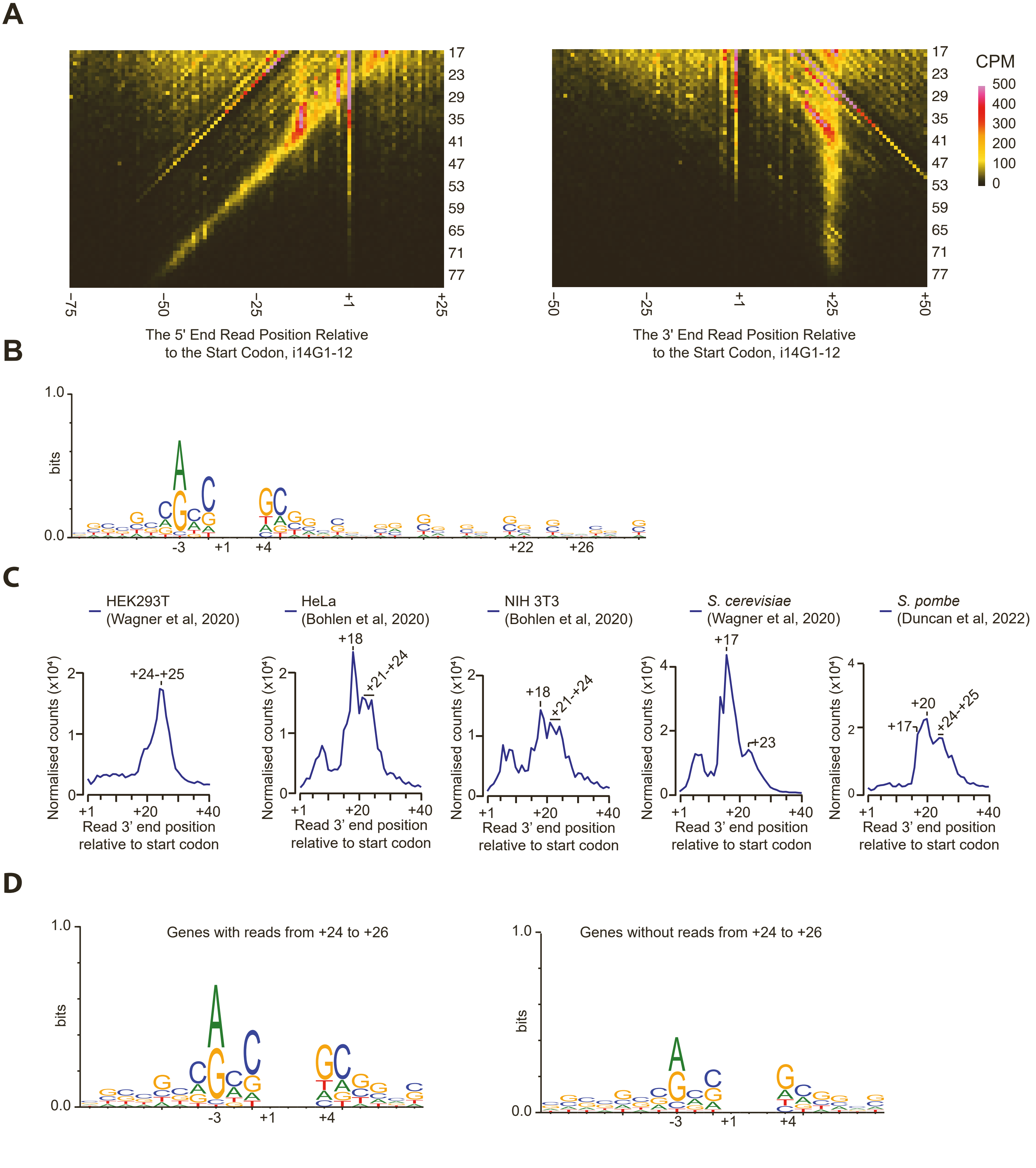

Figure S4: Footprint analyses on start codon, Related to Figure 4

A. Matrices of the read length versus the relative position of the 5' ends (left) or the 3' ends (right) read to start codon from cells treated with i14G1-12. The sum of normalised counts per million of the three replicates for each position and length is displayed.

B. Weblogo of the motif sequence around the start codong of genes enriched in reads ending at +24-+26 from DMSO treated cells.

C. Plots depicting the counts of the relative position of the 3' ends read to the start codon from 40S TCP-seq of different organisms.

D. Weblogo of the motif sequence around the start codong of genes with (left) or without (right) reads ending at +24-+26 from DMSO treated cells.

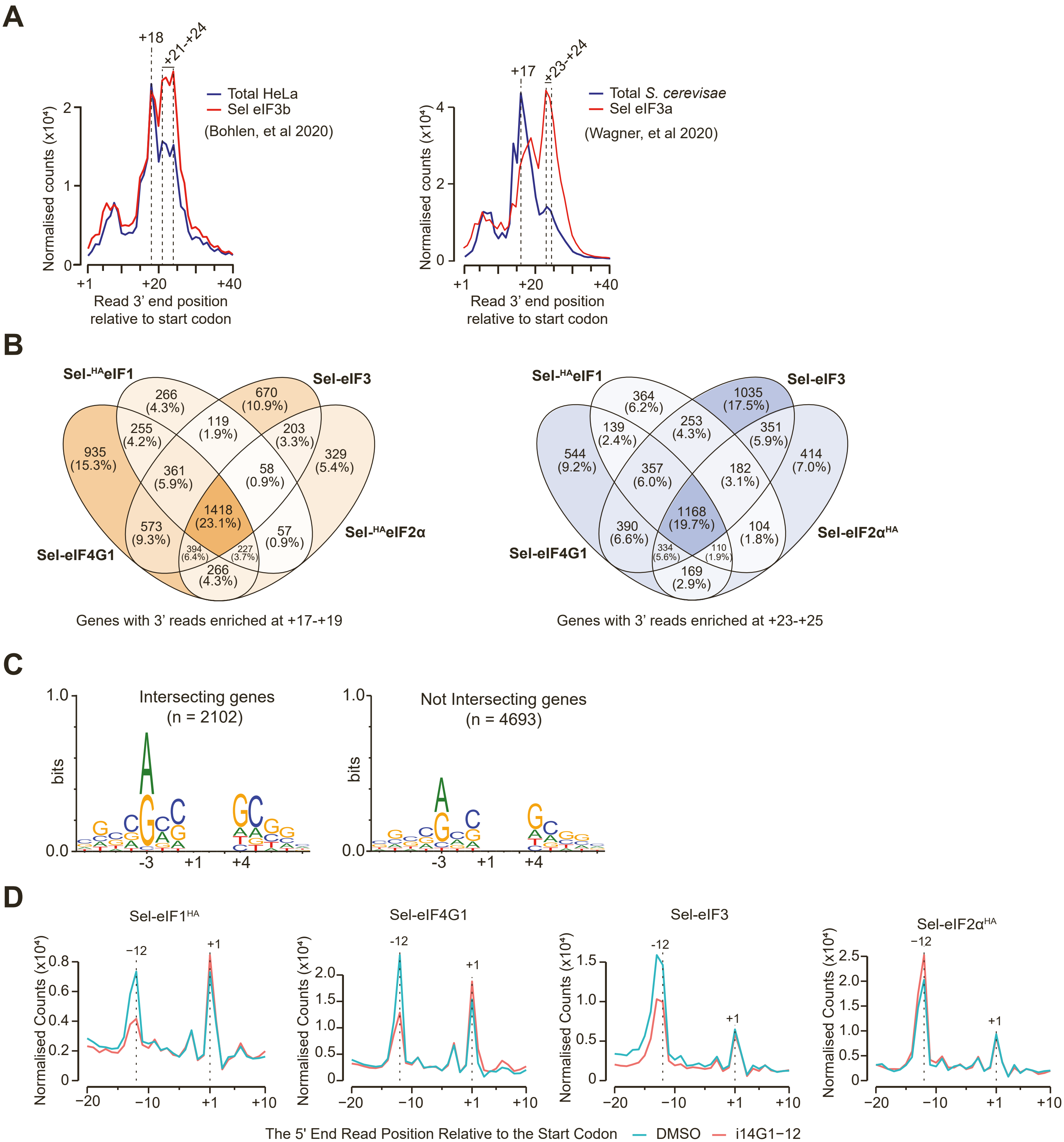

Figure S5: Footprint analyses on start codon from Selective TCP-seq data, Related to Figure 5

A. Plots depicting the counts of the relative position of the 3' end reads to the start codon from 40S TCP-seq (Total) or from Selective TCP-seq anti-eIF3b (left) or anti-eIF3a (right) from HeLa cells or *S. cerevisiae,* respectively.

B. Venn diagrams illustrating the overlapping genes that are enriched in reads ending at +17-+19 (left) or at +23-+25 (right) from the selective TCP-seq targeting eIF4G1, eIF1^{HA}, eIF3 or eIF2α^{HA}.

C. Weblogos showing the motif sequence around the start codon of intersecting (left) or not intersecting (right) enriched genes between the TCP-seq targeting eIF4G1, eIF1^{HA}, eIF3 and eIF2α^{HA} from Figure 5C.

D. Plots depicting the counts of the relative position of the 5' end reads to the start codon from the selective TCP-seq targeting eIF1^{HA}, eIF4G1, eIF3 or eIF2α^{HA}.